Building Data Apps In Python With Streamlit

Session 2: Working with User Input

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Exercise: What User Input?

At the start of the first session I asked you:

- What do you want to build after completing this course?

Now type into the chat:

- What input do you want your app to get from the user?
- What do you want your app to do with that input?

Course Agenda
(Big Picture)

1. Intro
2. Working with User Input

The key thing with data apps is that they get input from the user and then do something with it. That's what separates it from a Jupyter Notebook. In this section we'll learn how to let the user select a state, and then update the dataframe table to show information on just that state. We'll be using this skill in the next section as well.

Please Open

- 2-input/input.ipynb
- 2-input/input_app.py

In the terminal, please also change your directory to 2-input

If you are not using VS Code to view the notebook, type:

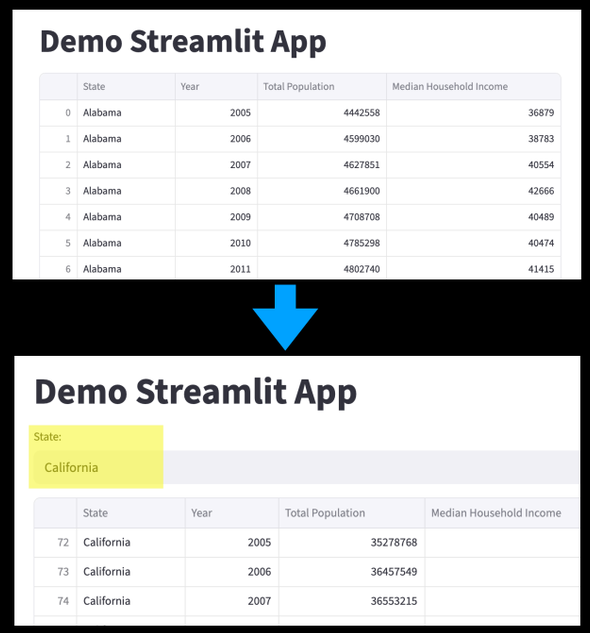
```
uv run jupyter notebook input.ipynb
```

Wait until everyone says they have done it.

The notebook for this section is very similar to my slides - please follow along.

"Select Box"

- Let the user "Select" which state they want to see data on
- Filter the dataframe to just see that data
- Print the result

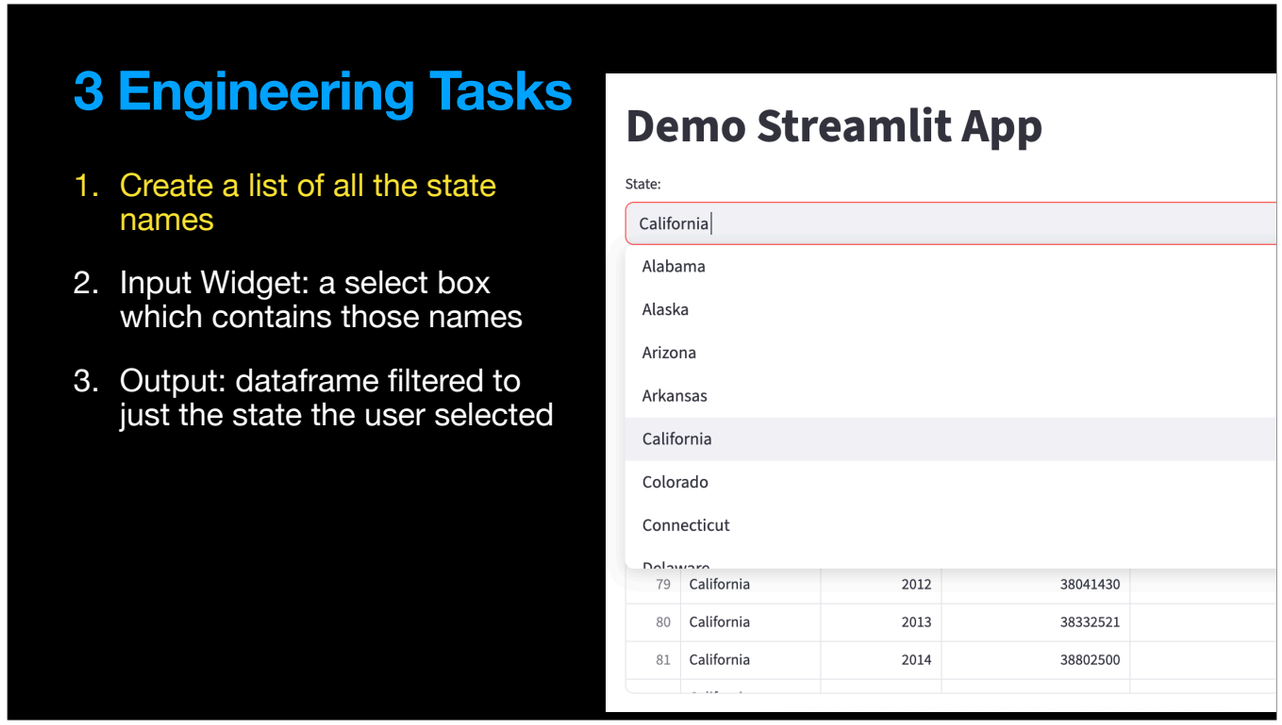


	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554
3	Alabama	2008	4661900	42666
4	Alabama	2009	4708708	40489
5	Alabama	2010	4785298	40474
6	Alabama	2011	4802740	41415

	State	Year	Total Population	Median Household Income
72	California	2005	35278768	
73	California	2006	36457549	
74	California	2007	36553215	

I want the user to be able to see data on just one state. To do this, we'll add what's called a "select box". Think of it like this: "the user gets to 'select' which state to view data on." We define the possible selections ahead of time.

In the first image, the table has data on all states. In the second image, the user selects a state, and the dataframe shows data on just that state.

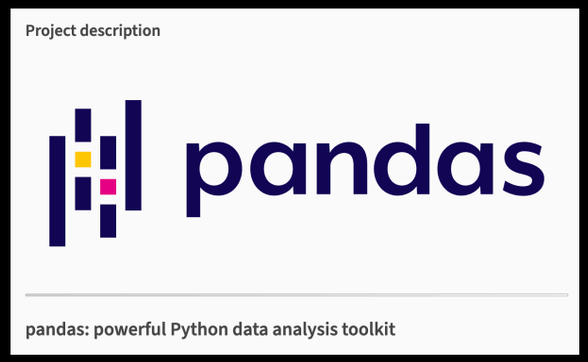


This feature (letting the user filter the dataframe to just one state) actually has 3 separate sub-tasks. When you're working on your own projects, it's helpful to break down tasks in this way as well. It normally takes some experience to get a knack for how to do this.

Let's now focus on the first task - getting a list of all the state names

pandas

- Tools for working with tabular data (i.e. csv files)
- "DataFrame" library



In order to get the list of the states in the data frame, we're going to use the pandas library. Pandas is the most popular library for working with tabular data in Python. I expect that most people in this course have prior experience with Pandas. But I want the course to be accessible to people who are new or rusty with it as well. So I'll give a short introduction / refresher

What is a DataFrame?

- A "Rectangle of Data"
- A "collection of columns"
- Each column
  - has a name
  - has values of the same type (text, number, boolean)
  - has the same length

	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554
3	Alabama	2008	4661900	42666
4	Alabama	2009	4708708	40489
...	...	...	...	...
931	Wyoming	2018	577737	61584
932	Wyoming	2019	578759	65003
933	Wyoming	2021	578803	65204
934	Wyoming	2022	581381	70042
935	Wyoming	2023	584057	72415

936 rows × 4 columns

```
type(df)
```

✓ 0.0s

```
pandas.core.frame.DataFrame
```

It can be helpful to think of a dataframe as "a rectangle of data". More precise is "a collection of columns, each of the same length".

-Each column has a name.

-The values in each column are of the same type

Each Column is a "Series"

- "Bracket notation"
- You can get "just" a column by typing **df[<column_name>]**
- Column name needs to be in quotes
- Each column is a "Series"

```
type(df['State'])
```

✓ 0.0s

```
pandas.core.series.Series
```

```
df['State']
```

✓ 0.0s

```
0      Alabama
1      Alabama
2      Alabama
3      Alabama
4      Alabama
        ...
931    Wyoming
932    Wyoming
933    Wyoming
934    Wyoming
935    Wyoming
Name: State, Length: 936, dtype: object
```

Working with "just" a column of a dataframe is very common. Pandas lets you do this with "bracket notation" - put the column name, in brackets, after the dataframe. The column name needs to be in quotes. You get the data in just the column back. The resulting object is of type "Series".

Series Methods

- .unique()
- .min()
- .max()
- ...

```
df['State'].unique()
```

✓ 0.0s

```
array(['Alabama', 'Alaska', 'Arizona', 'Arka
       'Colorado', 'Connecticut', 'Delaware
       'Florida', 'Georgia', 'Hawaii', 'Idah
       'Iowa', 'Kansas', 'Kentucky', 'Louisi
       'Massachusetts', 'Michigan', 'Minnesc
       'Missouri', 'Montana', 'Nebraska', 'N
       'New Jersey', 'New Mexico', 'New York
       'North Dakota', 'Ohio', 'Oklahoma',
       'Puerto Rico', 'Rhode Island', 'South
       'Tennessee', 'Texas', 'Utah', 'Vermor
       'West Virginia', 'Wisconsin', 'Wyomir
```

The Series class has a lot of useful methods. This is one reason why pandas is such a popular library - it makes common operations on dataframes easy. The method we want here is called unique(), and it returns the unique values in the series. It's perfect for letting the user select a state from our dataframe.

Exercise 2.1: Series Methods

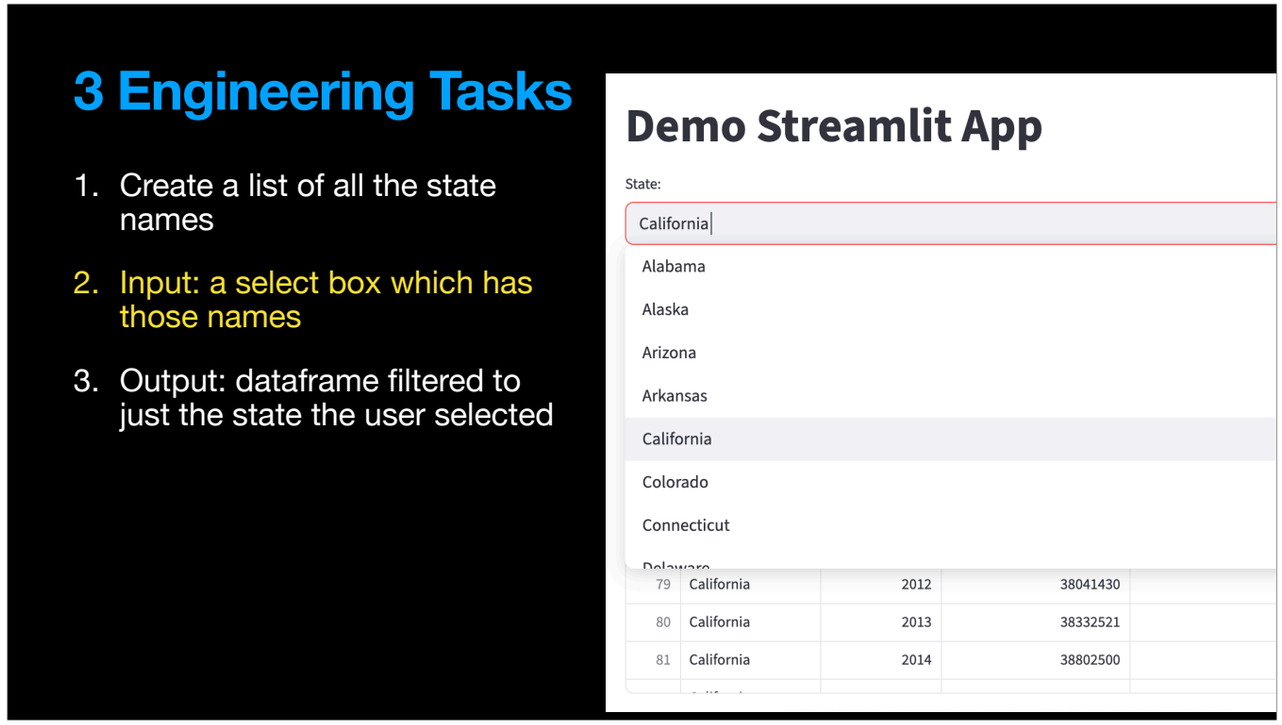
Start by executing all the cells in the notebook above this one. Then:

1. Calculate the unique values in the `Year` column. Does anything surprise you?
2. Use the `min()` and `max()` methods to calculate the min and max population in the dataset. Does anything surprise you?

Create new cells below this one as needed.

Working with Series is so important that I've created 2 exercises for you to work on.

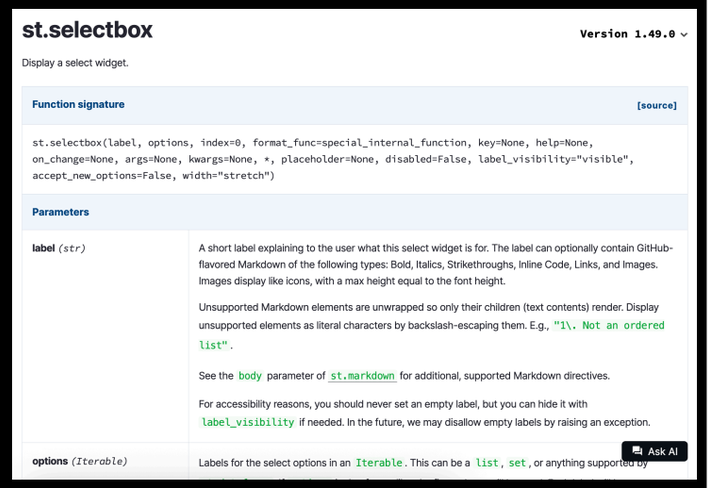
5 minutes



Now let's figure out how to get this select box into the app. This itself actually has two distinct subtasks. The first is to get any dummy selectbox into the app. The second is to update that dummy select box to have the right label (the text that says "State:") and the right values. There are two ways to learn do this: Google and an LLM.

Google

- Search: "*How do I create a select box in Streamlit?*"
- Wind up on the official documentation for st.selectbox
- Useful to be able to read. But a lot of info we don't need for our first box!
  - 15 parameters!



st.selectbox

Version 1.49.0

Display a select widget.

Function signature [source]

```
st.selectbox(label, options, index=0, format_func=special_internal_function, key=None, help=None,
on_change=None, args=None, kwargs=None, *, placeholder=None, disabled=False, label_visibility="visible",
accept_new_options=False, width="stretch")
```

Parameters

label *(str)*

A short label explaining to the user what this select widget is for. The label can optionally contain GitHub-flavored Markdown of the following types: Bold, Italics, Strikethroughs, Inline Code, Links, and Images. Images display like icons, with a max height equal to the font height.

Unsupported Markdown elements are unwrapped so only their children (text contents) render. Display unsupported elements as literal characters by backslash-escaping them. E.g., `"1\. Not an ordered list"`.

See the `body` parameter of `st.markdown` for additional, supported Markdown directives.

For accessibility reasons, you should never set an empty label, but you can hide it with `label_visibility` if needed. In the future, we may disallow empty labels by raising an exception.

options *(Iterable)*

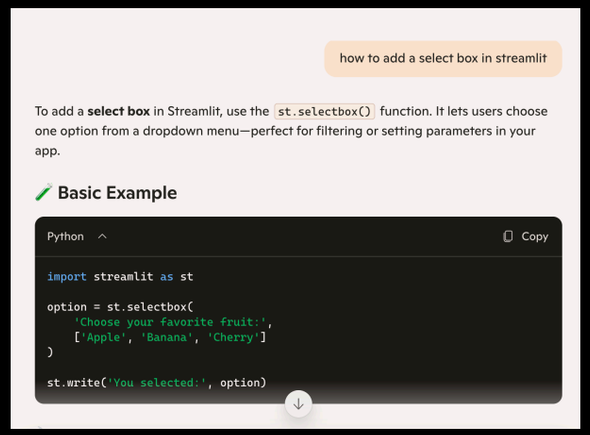
Labels for the select options in an `Iterable`. This can be a `list`, `set`, or anything supported by

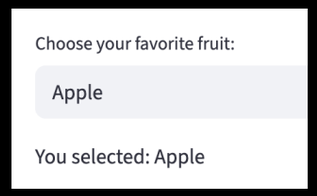
Ask AI

Take a minute to google "How do I create a select box in Streamlit?" Prior to LLMs I would often do search like this. You typically wind up on the official documentation page for the function (here st.selectbox). Learning how to navigate pages like this is a useful skill to gain. But it can be overkill for a beginner. It's most useful for understanding all the options.

LLM

- Visit copilot.microsoft.com
- Prompt: "*How do I create a select box in Streamlit?*"
- Response: copy-paste sample code
  - Only has the two required parameters





Now visit copilot.microsoft.com and ask the exact same question. What you get is a short paragraph and a code sample. Note that the code sample only has the two required parameters.

Exercise: input_app.py

- The file 2-input/input_app.py has the Streamlit example code for select boxes added to the app.
- Run the app with streamlit run input_app.py
- Update it to:
  - Let the user select from the unique states
  - Have a label that says something related to the options

how to add a select box in streamlit

To add a **select box** in Streamlit, use the `st.selectbox()` function. It lets users choose one option from a dropdown menu—perfect for filtering or setting parameters in your app.

Basic Example

Python Copy

```python
import streamlit as st

option = st.selectbox(
    'Choose your favorite fruit:',
    ['Apple', 'Banana', 'Cherry']
)

st.write('You selected:', option)
```

Choose your favorite fruit:

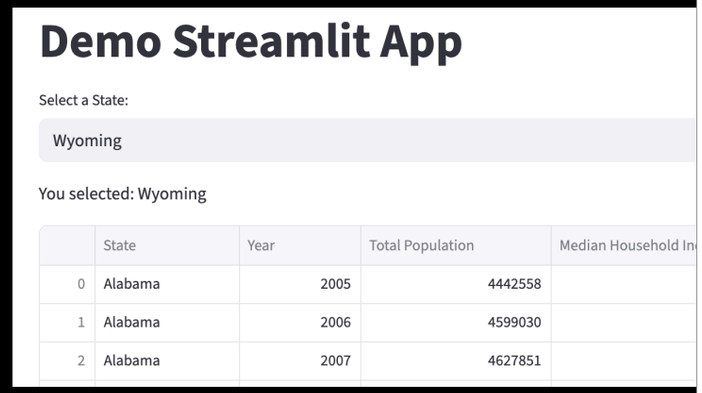
Apple

You selected: Apple

10 minutes

3 Engineering Tasks

1. Create a list of all the state names
2. Input: a select box which has those names
3. Output: dataframe filtered to just the state the user selected



Right now we have a perfectly functioning select box. But we're still displaying the full dataframe. For example, I selected "Wyoming" but the dataframe still has rows for Alabama. To fix this, we'll have to use what Pandas calls "Boolean Indexing".

Boolean Indexing

- Filter rows based on a condition
- First create a boolean Series
- Then feed the series to the dataframe using [] notation
- Example: Show me rows where the population is < 1M

	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554
3	Alabama	2008	4661900	42666
4	Alabama	2009	4708708	40489
...	...	...	...	...
931	Wyoming	2018	577737	61584
932	Wyoming	2019	578759	65003
933	Wyoming	2021	578803	65204
934	Wyoming	2022	581381	70042
935	Wyoming	2023	584057	72415

936 rows × 4 columns

Boolean indexing lets you filter rows in a DataFrame based on a condition. You create a Series of True and False values—called a Boolean mask—and use it to select only the rows where the condition is True.

I'll walk you through an example of selecting rows that have a population less than 1 million. Then I'll give you an exercise where you select rows with a particular state.

Note that Alabama has a population > 1M, and Wyoming has a population < 1M

Column / Series

- We care about the Total Population column / series
- We want to create a new Series where each element is
  - True if the element is < 1M
  - False otherwise

```
df['Total Population']
```

✓ 0.0s

```
0      4442558
1      4599030
2      4627851
3      4661900
4      4708708
        ...
931     577737
932     578759
933     578803
934     581381
935     584057
Name: Total Population, Length: 936, dtype: int64
```

Vectorization

- Pandas lets you vectorize a test over a Series
- This applies the test to each element of the Series
- The result is a boolean Series

```
df['Total Population'] < 1000000
```

✓ 0.0s

```
0      False
1      False
2      False
3      False
4      False
       ...
931     True
932     True
933     True
934     True
935     True
Name: Total Population, Length: 936, dtype: bool
```

Boolean Indexing

- Put a Boolean Series inside [] after a dataframe
- Only rows that are True are returned
- First element of the series corresponds to the first row of the dataframe.

```
df[df['Total Population'] < 1000000]
```

✓ 0.0s

	State	Year	Total Population	Me
18	Alaska	2005	641724	
19	Alaska	2006	670053	
20	Alaska	2007	683478	
21	Alaska	2008	686293	
22	Alaska	2009	698473	
...	...	...	...	
931	Wyoming	2018	577737	
932	Wyoming	2019	578759	

You can read it as "df where the total population column is less than 1M"

Confusing Notation

- [] is being used twice
- Inner []:
  - df['Total Population']
  - Conceptually: Column extraction
  - Given a string
  - Returns a Series
- Outer []:
  - df[<Boolean Series>]
  - Conceptually: Row filtering
  - Given a Boolean Series
  - Returns a dataframe

```
df[df['Total Population'] < 1000000]
```

✓ 0.0s

	State	Year	Total Population	Me
18	Alaska	2005	641724	
19	Alaska	2006	670053	
20	Alaska	2007	683478	
21	Alaska	2008	686293	
22	Alaska	2009	698473	
...	...	...	...	
931	Wyoming	2018	577737	
932	Wyoming	2019	578759	

Sometimes people find Boolean Indexing to be confusing. I think this is because on a single line, the bracket notation is being used twice. And each time, it's being used in a completely different way.

2-line Option

- Put the creation of the Boolean Series on its own line
- Convention: call the Series mask.

```
mask = df['Total Population'] <= 1000000
df[mask]
```

✓ 0.0s

	State	Year	Total Population	Median Househ
18	Alaska	2005	641724	
19	Alaska	2006	670053	
20	Alaska	2007	683478	
21	Alaska	2008	686293	
22	Alaska	2009	698473	
...	...	...	...	
931	Wyoming	2018	577737	
932	Wyoming	2019	578759	
933	Wyoming	2021	578803	
934	Wyoming	2022	581381	
935	Wyoming	2023	584057	

130 rows × 4 columns

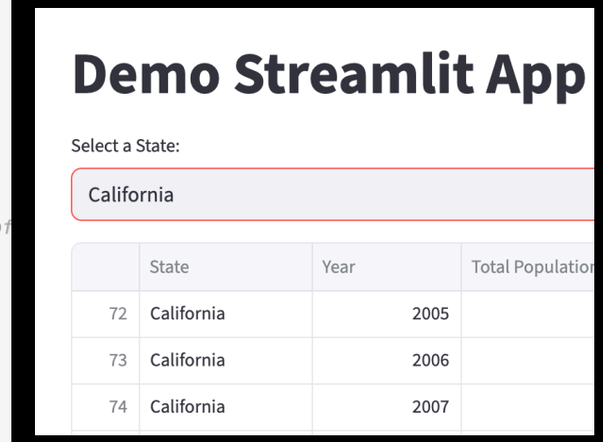
Exercise 2.2: Boolean Indexing

1. Create a variable called `mask` that says which rows in `df` are in California. Use it to subset the dataframe.
2. Update your app to filter the dataframe to show rows in the state the user selected.

Use masking in a Jupyter Notebook to subset which rows in the dataframe are in California.
Then update the app to subset the dataframe based based on the state the user selected.
10 minutes

session2 > input_app.py > ...

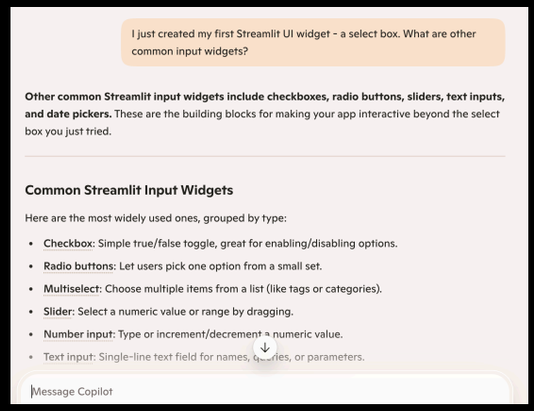
```python
import streamlit as st
import pandas as pd

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

# Exercise: Change this code to:
# 1. Ask the user to select a state
# 2. Have it populate with all the list of
option = st.selectbox(
    'Select a State:',
    df['State'].unique()
)

mask = df['State'] == option
df = df[mask]

st.dataframe(df)
```



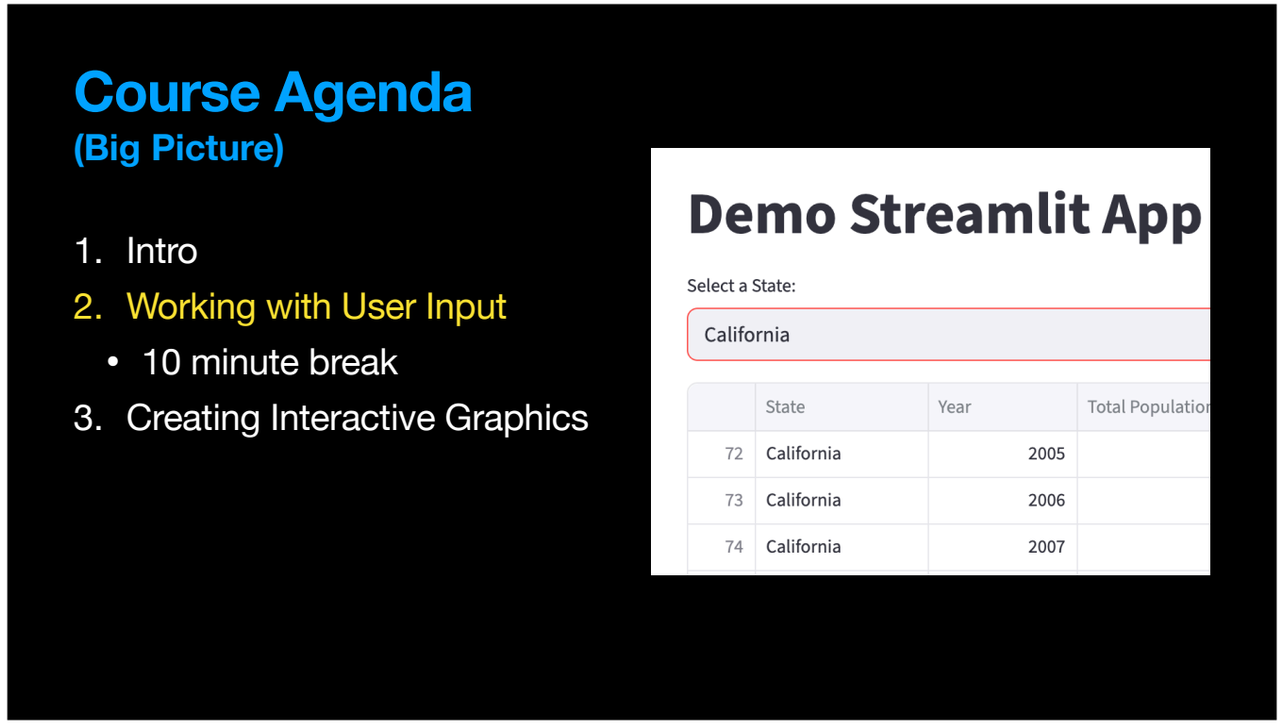
My solution

Exercise: Copilot & Input Widgets

- Goal: Use Copilot to help you learn more about Streamlit UI Widgets.
- Prompt #1: "*I just created my first Streamlit UI widget - a select box. What are other common input widgets?*"
- Pick a widget you want to learn more about.
- Prompt #2: "*Show me an example of <widget>*"
- Copy and paste the example code into your app. Click "Rerun".



One of my goals in this course is to help you learn how to use Copilot during development. In this course the only input widget we'll be using are Select Boxes. But Streamlit has dozens more ways to get input. Ask Copilot for other popular widgets. Then select one that interests you. Then do a followup prompt asking for an example of how to use the one that interests you. Then add it to input_app.py and click Rerun. Give URL: copilot.microsoft.com. 10 minutes.



We now have an app that lets users filter a dataframe. In the next section we'll learn how to add graphs to the app - for example, how to graph the population of the selected state over time. Before doing that, though, let's take a 10 minute break.

But first: are there any questions about working with user input in Streamlit apps?